



Tanushka Das

Roll no:- 240123066

B.Tech- Mathematics And Computing

Indian Institute Of Technology, Guwahati

+91-8250877321

dastanushka06@gmail.com

d.tanushka@iitg.ac.in

github.com/tanushka509

linkedin.com/tanushka-das-4b3b9735b

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.10(Current)	2024-Present
Senior Secondary	Telangana State Board of Intermediate Education	98.7%	2024
Secondary	Board Of Secondary Education,Telangana	10.00	2022

PROJECTS

- Corporate Transportation Optimization System (Vehicle Routing Engine)** *Januray 2026 - March2026*
Kriti-inter hostel Technical Competition *GitHub*
 - **Algorithm:**Built a scalable optimization engine using K-Means clustering, greedy insertion, repair heuristics, and 2-opt local search to solve a constrained vehicle routing problem.
 - **Contribution:** Co-designed and implemented the core backend logic in Python, handling real-time assignment, constraint validation (time windows, capacity, preferences), and route optimization.
 - **Impact and Learning:** Applied algorithm design, data structures, and performance optimization to deliver an end-to-end system achieving 81–95 percent cost reduction with efficient API integration (FastAPI + Node.js).
- CVM++: Custom Compiler & Stack-Based Virtual Machine** *April 2026 - May 2026*
Coding Club *GitHub*
 - **Architecture:** Built a custom compiler and stack-based VM in C++, implementing the complete source-to-execution pipeline.
 - **Parsing:** Developed a handwritten lexer, recursive-descent parser, and AST generator with operator-precedence handling.
 - **Compilation:** Designed a custom bytecode format and compiler backend for code generation.
 - **Execution:** Implemented a stack-based bytecode interpreter with variable storage and expression evaluation.
- AI powered Resume Analyzer and Job tracking Application** *April 2026 - May 2026*
Coding Club *GitHub*
 - **Architecture:** Built a custom compiler and stack-based VM in C++, implementing the complete source-to-execution pipeline.
 - **Parsing:** Developed a handwritten lexer, recursive-descent parser, and AST generator with operator-precedence handling.
 - **Compilation:** Designed a custom bytecode format and compiler backend for code generation.
- DISRUPT Startup Bootcamp Website** *July. 2025*
Ecell IIT Guwahati *Website | GitHub*
 - Collaborated with the WebOps team to develop and maintain the **official DISRUPT startup bootcamp website** for E-Cell IIT Guwahati.
 - Built frontend features and improved user experience for a production platform serving **70+** startups and **25+** mentors.

TECHNICAL SKILLS

- Programming:** C, C++,JavaScript, Java*
- Web Technology:** HTML,CSS,GSAP*, React ,Node Js, Express Js
- Databases:** MongoDB, MYSQL
- Tools/Frameworks:** Git, Github, REST APIs, Postman
- Operating Systems:** Windows, Linux*

* Elementary proficiency

KEY COURSES TAKEN

- Mathematics:** Linear Algebra, Basic Calculus, Discrete Maths, Probability & Random Processes, Real analysis,Design and Analysis of Algorithms,Financial Engineering*,Data structures and Algorithms
- Computer Science:** Digital Design, Database Management System, Computer Organization and Architecture **Awarded AA grade*

ACHIEVEMENTS

- JEE ADVANCED**,Ranked in the top 2% among 180,000 candidates nationwide *2024*
- JEE MAINS**,,Ranked in the top 0.5% among 1,000,000 candidates nationwide *2024*
- Secured 1st position in the "Zero to One"**,in E-Cell IIT Guwahati's "Zero to One" event by developing a startup solution for Zyp Electric. *2025*
- Goldman Sachs Engineering Hackathon (CS Domain)**,Scored 245/300 solving a drone fleet optimization problem with No-Fly Zone (NFZ) constraints, alongside algorithmic and software engineering challenges. *2025*

POSITIONS OF RESPONSIBILITY

- Senior Executive Disrupt,ECell Club, IIT Guwahati** *May 2025-Jan 2026*
- Public Relations Associate**,Alcheringa, IIT Guwahati *Nov. 2024 - Mar-2025*